

STUDIES FOR A NOVEL GENERATION OF THE BEAM PROFILE MEASUREMENT (BEAM WIRE SCANNER) FOR THE LARGE HADRON COLLIDER (LHC)

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Abstract

Aiming for the improvement of the reliability and maintainability of the LHC linear wire scanners, a completely new concept is currently under development. The innovative design will eliminate the use of bellows, feature a more precise and optimized wire positioning system, and incorporate a significantly more robust and resilient mechanical structure. Four next-generation instruments are scheduled for installation in the LHC during Long Shutdown 3, enabling both horizontal and vertical beam profiling for beam 1 and beam 2 with improved accuracy and stability. This paper presents the conceptual study, detailed mechanical design, integration strategy, and the initial experimental tests of the newly developed magnetically driven linear wire scanner.

INTRODUCTION

In the Large Hadron Collider (LHC), Wire Scanners (BWS) play a key role as the reference instrument for machine developments and are used to calibrate other profile measurement devices which provide precise bunch-by-bunch measurements at all energy levels [1].

The scanners currently installed in the LHC were inherited from the Large Electron-Positron (LEP) Collider after its decommissioning, with the original design dating back to the 1980s. In recent years, several reliability issues have required interventions, including:

- Bellow leakage (four cases, most recently in 2023)
- Fork blockage (2015)
- Loss of wire tension (two scanners, 2018)

In 2015, a consolidation project was launched to improve the driving system and the external mechanics. As part of this work, four scanners were upgraded to “hybrid” systems by replacing the legacy motion mechanism with a modern ball screw drive (Fig. 1). These new linear drives were integrated into the standardised LHC Injectors Upgrade (LIU) control system [2].

To date, the in-vacuum mechanics remain unmodified since installation, and are not optimised for positional accuracy and vibration reduction. Furthermore, the continued use of bellows to enable in-vacuum motion has led to ongoing reliability concerns. It was therefore decided to proceed with a complete mechanical redesign as part of a second stage of the consolidation, both to mitigate the need for future interventions and to improve measurement accuracy.

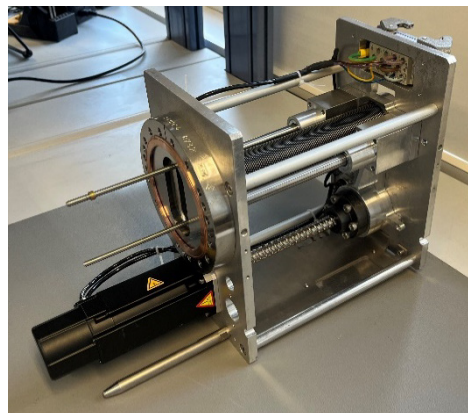


Figure 1: Hybrid linear wire scanner.

The large time span since the legacy design has provided an opportunity to integrate modern materials and motion technologies. This new mechanical overhaul project began in November 2023. A redesigned scanner has since been developed, and the first prototype is now in production for installation during the LHC year-end technical stop (YETS) 2025.

The implementation plan foresees the installation of a prototype during the winter of 2025/26 to validate the design. Full replacement of the eight existing instruments is planned in two stages: the first during Long Shutdown 3 (2026–2030), followed by the second during Operation Run 4 (2030–2033).

This paper presents the conceptual study, detailed mechanical design, integration strategy, and initial experimental tests of a newly developed magnetically driven linear wire scanner.

SYSTEM OVERVIEW

During the conceptual design stage the key constraints and requirements of the scanner were identified:

- The design phase was dedicated to the development of a new instrument compatible with the existing control system and electronics designed for the (BWS) installed in CERN’s three circular injectors.
- The design had to comply with the stringent vacuum specifications of the LHC while ensuring high mechanical reliability through a bellows-free architecture.
- A key design driver was impedance optimization, aiming to maintain or improve the overall longitudinal and transverse impedance of the LHC.

- Significant effort was devoted to optimizing the instrument's integration within the constrained environment of the LHC tunnel, with particular emphasis on alignment, accessibility, and maintainability.

Design choices were strongly influenced by considerations related to installation efficiency, handling, and operational robustness.

MECHANICAL DESIGN

The newly developed instrument (Fig. 2) consists of two principal assemblies. The first is the vacuum chamber incorporating impedance mitigation elements. The second is the measurement module, which can be detached for servicing and consists of:

- An external actuation system with magnetic coupling to ensure reliable motion transmission without compromising vacuum integrity.
- A high-resolution optical encoder for precise position feedback.
- A ceramic sliding rail providing low-friction, wear-resistant guidance,
- A structural support frame accommodating the carbon wire is employed for high-accuracy beam profile measurements.

To meet the stringent requirements for vacuum compatibility, impedance minimization, and long-term operational reliability, the design effort prioritized a highly compact and mechanically integrated architecture from the outset.

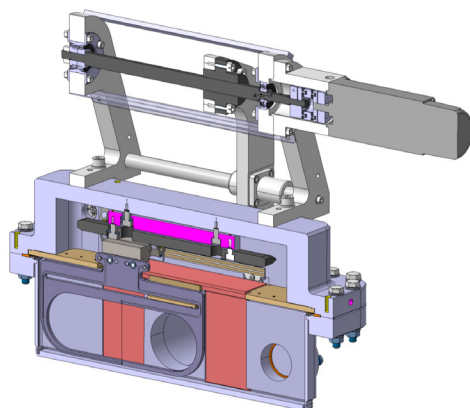


Figure 2: Linear wire scanner external motion system and in vacuum motion system.

Optical encoder design (Fig. 3): is derived from the system implemented on the LIU rotary wire scanners [1]). In this application, the traditional glass disk is replaced by a precision glass scale onto which the identical optical pattern is deposited, enabling high-resolution position measurement in a linear configuration.

Motion transmission system (Fig. 3): employs a magnetic coupling in combination with a ceramic linear guide incorporating hybrid ball bearings, replacing the previous bellows-based mechanical assembly (Fig. 3). This approach significantly enhances mechanical robustness while virtually eliminating the risk of vacuum breaches [3].

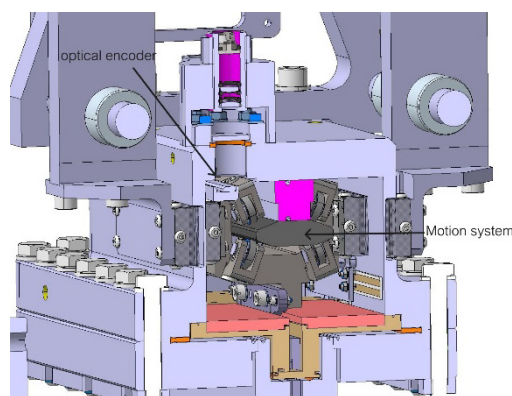


Figure 3: Linear wire scanner optical encoder and permanent magnetic coupling.

Beam-coupling impedance mitigation (Figs. 4 and 5) represented a significant design challenge. Efforts were concentrated on displacing critical resonance frequencies outside the LHC's operational spectrum and minimizing power deposition on sensitive components. Key strategies included the reduction of beam-facing volumes through an optimized internal geometry, the creation of two isolated vacuum volumes to limit beam interaction, and the strategic placement of absorber tiles. These alumina tiles, coated with titanium as well as the RF contacts, effectively reduced parasitic modes. Electromagnetic simulations of the final configuration yielded highly favourable results, confirming the effectiveness of the adopted approach [4].

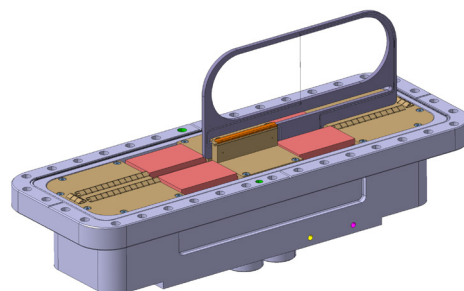


Figure 4: Linear wire scanner with absorbers tiles and RF contacts.

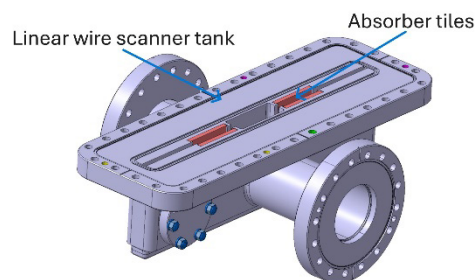


Figure 5: Linear wire scanner tank with absorber tiles.

The adopted design modifications enabled a significant attenuation of the RF modes and a frequency shift toward higher values, approaching or exceeding 1.5 GHz.

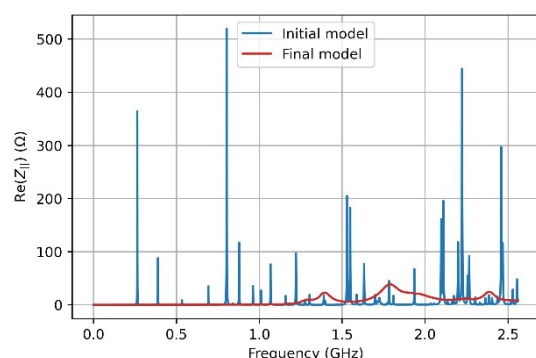


Figure 6: Impedance simulation before and after design optimisation.

Initial simulations indicated a power deposition of approximately 16 W on the instrument. Through the implemented design optimizations, this value was reduced to a range of 4–7 W (Fig. 6). The majority of this energy is expected to be absorbed by the dedicated absorber tiles, resulting in a maximum temperature of approximately 180 °C at the tiles and about 50 °C at the ceramic support frame. Both temperatures remain well below the operational and material limits, confirming that thermal load does not constitute a limiting factor for the instrument's performance or longevity (Fig. 7).

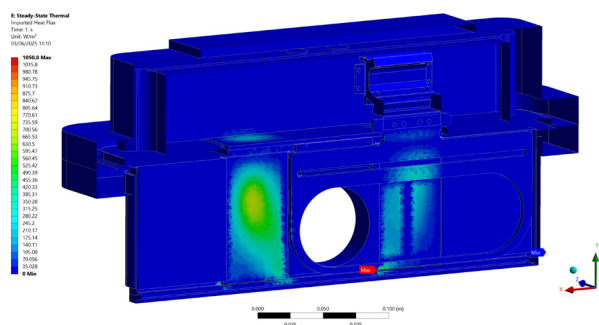


Figure 7: Power deposition simulation.

This component has undergone a complete redesign to optimize the balance between mechanical stiffness and mass. Alumina (Al_2O_3) was selected as the structural material due to its favorable mechanical properties and compatibility with stringent vacuum and impedance requirements.

The wire continuity monitoring system has likewise been re-engineered. The previous approach, based on a pair of insulated signal wires, posed potential reliability and impedance issues in the LHC environment. The new design employs a pantograph mechanism integrated into the frame, sliding on two Thick-film screen printed conductive tracks on a ceramic substrate (Fig. 8). This configuration enables reliable continuity measurements while minimizing impedance impact and significantly reducing the risk of signal loss during operation.

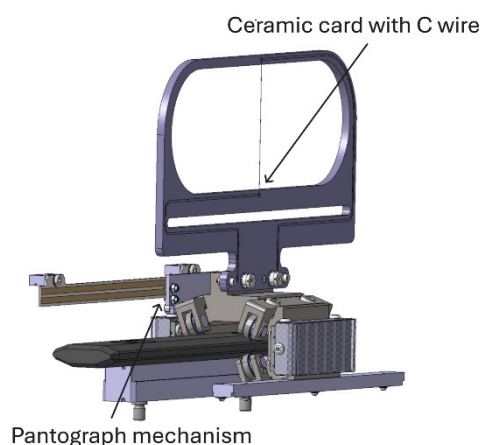


Figure 8: Equipped sliding card.

FABRICATION AND TEST

A test rig for the newly developed linear wire scanner components, including the magnetic coupling and ceramic linear guide, has been constructed. Experimental results confirmed the design choices made during the development phase. Reliability testing demonstrated that the system could perform 80,000 measurements without exhibiting any noticeable degradation, thereby validating the fabrication of two prototype units.

The first prototype is scheduled for installation in the LHC during the upcoming winter shutdown (YETS 2025/26) to undergo beamline testing during the 2026 run. The second prototype will remain in the laboratory for continued optimization of the control system, magnetic coupling, and overall instrument performance.

CONCLUSION

The new generation of beam wire scanners for the LHC is designed to deliver enhanced performance, including higher positional accuracy, reduced vibration, and extended operational lifetime. The design also emphasizes simplified assembly and maintenance, thereby reducing both installation time and operational complexity within the LHC tunnel. These improvements collectively provide a robust transverse profile measurement system capable of addressing the stringent requirements of the High-Luminosity LHC era, scheduled to commence in 2030. The installation program, as planned by the Beam Instrumentation Group—comprising four instruments during Long Shutdown 3 and an additional four during Run 4—will be validated in the second half of 2026, following the evaluation of the two prototypes currently under fabrication.

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